

Difference-of-Convex Regularization for Graph Learning by Differentiable Programming

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Abstract—Laplacian-regularized minimization is fundamental in signal processing and machine learning, but is limited by the dense and ill-conditioned nature of the graph Laplacian pseudoinverse. While the Laplacian itself is sparse, its pseudoinverse is dense and often ill-conditioned, rendering direct computation impractical at scale. Moreover, pseudoinverse learning is more challenging than Laplacian learning. To address this challenge, this paper considers the setting where the graph Laplacian is given and proposes a Difference-of-Convex Regularizer (DCR) graph learning framework that approximates the spectral action of the Laplacian pseudoinverse without direct inversion via regularized Maximum Likelihood Estimation (MLE). By reformulating Laplacian-Regularized Nonnegative Least Squares (LR-NNLS) through a dual representation, DCR decouples pseudoinverse learning from instance-specific inference and enables efficient primal solution reconstruction via a differentiable dual-guided learning scheme. We establish theoretical guarantees on stability and the existence of a unique fixed point for DCR algorithm. Numerical experiments demonstrate improved performance over convex solvers and graph filtering baselines and robust performance across diverse graph topologies.

Index Terms—Laplacian-regularized optimization, Graph learning, Toland duality, Difference-of-convex, Differentiable programming.

I. INTRODUCTION

Laplacian-regularized minimization [1] is a fundamental optimization paradigm that integrates task-specific objectives with graph-based smoothness priors, enabling learning and estimation to exploit graph structure rather than treating variables independently [2]. It has been widely used in machine learning, computer vision, and statistical estimation. A broad class of Laplacian-Regularized Minimization Problems (LRMPs) can be written as:

$$\min_{x \in \mathbb{R}^n} f(x) + \frac{1}{2} x^\top L x, \quad (1)$$

where f is convex and $L \succeq 0$ is a symmetric, singular graph Laplacian with $L\mathbf{1} = 0$, promoting smoothness by penalizing discrepancies across graph-connected components [1].

In many real-world applications, LRMPs arise with intrinsic constraints. The variables often represent physical or semantic quantities such as abundances, activation coefficients, importance scores, or resource allocations that are inherently non-negative. This naturally leads to constrained formulations that enforce graph-induced smoothness while preserving feasibility

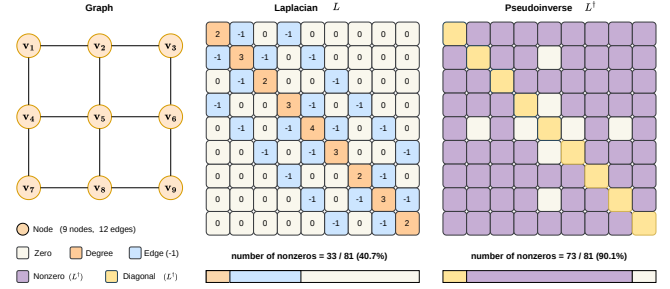


Fig. 1. While the graph Laplacian L is sparse, its pseudoinverse $L^\dagger$ is dense, rendering direct computation memory- and compute-intensive at scale.

and interpretability. A representative example is the Laplacian-Regularized Nonnegative Least-Squares (LR-NNLS) problem:

$$\min_{x \succeq 0} \frac{1}{2} \|Ax - b\|_2^2 + \frac{1}{2} x^\top L x, \quad (2)$$

which is widely used in applications such as semi-supervised ranking on graphs [3] and graph-guided nonnegative coding [4]. These ideas can also be extended to matrix problems, such as hyperspectral unmixing [5], graph-regularized nonnegative matrix factorization [6], and Laplacian-regularized nonnegative representation [7]. In all these cases, the nonnegativity constraint plays a key role in defining the solution space and influencing how the problem is solved.

While the Laplacian term enforces smoothness across connected nodes, the singularity of L implies that classical LRMP solutions are characterized through the Moore–Penrose pseudoinverse $L^\dagger$ [8]. In what follows, we refer to $L^\dagger$ as the Laplacian pseudoinverse. As shown in Figure 1, although L is sparse, its pseudoinverse is dense and often ill-conditioned, resulting in $\mathcal{O}(n^3)$ time complexity and $\mathcal{O}(n^2)$ memory complexity for large graphs [9]. This challenge is further exacerbated under additional constraints, such as nonnegativity in (2), which preclude closed-form solutions and require iterative or implicit solvers. More broadly, the difficulty of manipulating dense inverse operators constitutes a recurring bottleneck in modern learning systems [10], where high-dimensional graph or kernel-induced mappings rarely admit direct computation.

Motivation. Typical approaches to LRMPs include projected gradient methods, which provide simple and scalable solutions under constraints without requiring explicit pseudoinverse computation. However, their convergence can be slow for ill-conditioned Laplacians, and they do not exploit the global spectral structure of the problem. Recent advances suggest

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replacing intractable operators with efficiently computable surrogates, such as spectral graph filtering and polynomial approximations (e.g., Chebyshev filtering) [11]–[13], a paradigm widely used in graph signal processing [14]–[16]. While these methods avoid explicit pseudoinverse computation, they rely on fixed spectral approximations rather than learning problem-adaptive inverse operators, and remain limited under nonnegativity constraints, leaving room for further improvement. This raises a central question: *Given Laplacian L , can we design a graph learning framework that approximates $L^\dagger$ in a problem-adaptive manner without direct computation, while naturally supporting nonnegativity constraints?*

Graph learning [17] often studies graph structures or graph-induced operators. In this work, we assume the Laplacian is given and focus on learning an approximation of its pseudoinverse, which often leads to a nonconvex problem due to its inherent complexity. Difference-of-Convex (DC) programming [18]–[21] with regularization, i.e., DC regularization [22]–[24], provides a principled way to incorporate structural priors through tractable formulations. *Toland duality* [25] further offers a useful framework for understanding such DC formulations from a nonconvex duality perspective. Moreover, regularized DC models allow better control of spectral properties while remaining efficiently solvable via the Convex-Concave Procedure (CCCP) [26]–[29]. This motivates learnable approximations of the graph Laplacian pseudoinverse that exploit the underlying graph structure and handle the resulting non-convex optimization via DC regularization and CCCP.

Contributions. To overcome the above limitations, we propose a structured graph learning framework for Laplacian-regularized optimization that replaces $L^\dagger$ with a learnable approximation via regularized Maximum Likelihood Estimation (MLE) and differentiable programming [30], enabling end-to-end optimization of complex programs through automatic differentiation [31]. Our main contributions are:

- We study the LR-NNLS problem (2), derive its dual formulation to decouple A from $L^\dagger$, and use Karush-Kuhn-Tucker (KKT) conditions to reveal the structure underlying primal solution recovery, motivating learnable and problem-adaptive pseudoinverse approximation.
- We propose a Difference-of-Convex Regularizer (DCR) graph learning framework that decouples pseudoinverse learning from instance-specific inference. A learned approximation is obtained via shrinkage-regularized CCCP iterations from regularized MLE, and primal solutions are recovered through dual-guided differentiable learning. The *novelty* of DCR lies in unifying learning-based pseudoinverse approximation, non-convex DC regularization with CCCP-based spectral control, and KKT-guided nonnegativity-aware reconstruction.
- We establish iteration stability and unique fixed-point guarantees for the DCR algorithm. Experiments across diverse graph topologies and scales demonstrate robustness across graph structures and clear gains over convex solvers and graph filtering baselines. The source code is publicly available.¹

II. RELATED WORK

A. Nonnegative LRMPs

Semi-Supervised Ranking. A closely related line of work is the semi-supervised ranking framework of [3], which studies large-scale graph-based ranking with rich node and edge metadata. The framework formulates ranking as an optimization problem over a nonnegative score vector, combining graph-based smoothness induced by random walks or graph Laplacians with explicit pairwise preference supervision. As a special case, the Laplacian rank formulation minimizes a quadratic objective defined by a directed graph Laplacian, subject to margin-based pairwise preference constraints and a nonnegativity requirement on the ranking vector. This yields a supervised Laplacian-regularized optimization problem that extends PageRank-style models and is closely related to the class of nonnegative LRMPs of our work.

Graph-Aware Dictionary Learning. A related example appears in the badge-based document representation framework of [4], which encodes each document as a nonnegative sparse coefficient vector over a learned badge dictionary. The coding stage solves a nonnegative least-squares/lasso-type problem with a graph-guided fused regularizer on a badge relation graph [32], [33], encouraging correlated badges to take similar coefficients via weighted pairwise differences. This yields a structured nonnegative optimization that combines reconstruction, sparsity, and graph-based smoothness. Although the regularizer is not a quadratic Laplacian form, it is closely related to nonnegative LRMPs in that it integrates nonnegativity with graph-structured regularization for coherent representations.

Hyperspectral Data Unmixing. Graph Laplacian regularization has also been studied in nonnegative matrix-valued optimization problems beyond ranking. In hyperspectral data unmixing, Ammanouil et al. [5] proposed a Laplacian-regularized convex formulation to estimate abundance maps under nonnegativity and sum-to-one constraints. The model combines a quadratic data-fidelity term with a graph-based smoothness regularizer, promoting spatial and spectral consistency over a similarity graph. This formulation can also be viewed as a nonnegative Laplacian-regularized least-squares problem, closely related to the nonnegative LRMPs considered in this work, differing mainly in its matrix-valued variables and application setting.

Nonnegative Matrix Factorization. Nonnegative Laplacian-regularized problems also arise in nonnegative matrix factorization. Graph regularized nonnegative matrix factorization [6] extends standard nonnegative matrix factorization by incorporating a Laplacian term to preserve data geometry, leading to a quadratic reconstruction loss with a smoothness penalty under elementwise nonnegativity constraints. It is a nonnegative Laplacian-regularized problem where the Laplacian enforces graph smoothness and feasibility is maintained via constrained updates. Although designed for matrix-valued representations in clustering and representation learning, its core structure aligns closely with the nonnegative Laplacian-regularized formulations.

Nonnegative Self-Representation. Nonnegative Laplacian-regularized formulations have also been studied in self-

¹Source code is available at <https://github.com/convexsoft/LRMP>.

representation models for clustering and dimensionality reduction. For instance, Laplacian-regularized nonnegative representation [7] learns a nonnegative coefficient matrix under a self-expressive framework, enhanced with graph-based smoothness regularization. This formulation promotes interpretability via nonnegativity while enforcing smoothness over a similarity graph. It can be viewed as a matrix-valued extension of nonnegative LRMPs, closely related to LR-NNLS, but differing in its self-representation structure and loss design.

B. Graph Learning

Laplacian / Graph Structure Learning. A major line of work focuses on learning the graph Laplacian. Pavez et al. [34] formulate this task as maximum likelihood estimation of a precision matrix under generalized Laplacian constraints, enabling flexible Gaussian Markov random field modeling. Data-driven approaches further construct graphs via structured representations, such as the nonnegative low-rank and sparse framework in [35], which captures global and local structures, and graph refinement methods like [36], which project an initial similarity matrix onto a nonnegative, low-rank, and positive semidefinite set to obtain a denoised graph.

Laplacian Pseudoinverse Learning. Complementary to structure learning, recent works aim to directly learn or approximate the Laplacian pseudoinverse, which encodes global connectivity and diffusion behavior. Alfke et al. [37] learn the pseudoinverse implicitly from graph signals via algebraic constraints, and Zhu et al. [38] develop structured approximations for large graphs, while Lu et al. [39] improve efficiency and stability via diagonal preconditioning. Our work focus on learning the pseudoinverse of a given Laplacian matrix using regularized MLE and CCCP.

III. LAPLACIAN-REGULARIZED NONNEGATIVE LEAST SQUARES

Given a data matrix $A \in \mathbb{R}^{m \times n}$ and a vector $b \in \mathbb{R}^m$, the objective is to obtain a nonnegative decision variable $x \in \mathbb{R}^n$ by solving:

$$\min_{x \geq 0} \frac{1}{2} \|Ax - b\|_2^2 + \frac{1}{2} x^\top Lx, \quad (3)$$

where $L \in \mathbb{R}^{n \times n}$ is given and denotes the graph Laplacian associated with an undirected weighted graph $(\mathcal{V}, \mathcal{E})$, where $w_{ij} \geq 0$ denotes the weight of edge (i, j) . The quadratic penalty admits:

$$x^\top Lx = \sum_{(i,j) \in \mathcal{E}} w_{ij} (x_i - x_j)^2. \quad (4)$$

Spectral Challenges. Assuming $L^\dagger$ is known, solving (3) remains challenging due to spectral ill-conditioning. As shown in Fig. 2, using a chain graph Laplacian ($n = 500$, $m = 25$, $w = 0.01$, $\kappa(A^\top A + L) \approx 3.64 \times 10^4$) as a representative ill-conditioned instance, although the objective gap decreases rapidly for the first-order methods such as projected gradient descent, $\|G_\eta(x)\|$ remains above 10^{-3} throughout 3000 iterations, indicating slow convergence to stationarity. The nonnegativity constraint further complicates the optimization.

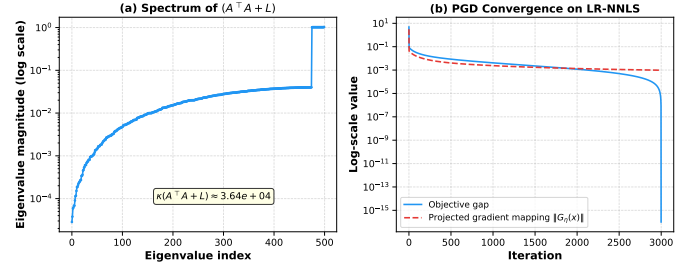


Fig. 2. Spectral and convergence properties of the LR-NNLS problem (3) ($n = 500$, $m = 25$, chain graph, $w = 0.01$, $\kappa(A^\top A + L) \approx 3.64 \times 10^4$). (a) Eigenvalue spectrum of $(A^\top A + L)$, showing severe ill-conditioning with smallest eigenvalue near 10^{-5} . (b) Although the objective gap decreases rapidly to near machine precision, the projected gradient mapping norm $\|G_\eta(x)\|$ remains above 10^{-3} throughout 3000 iterations, indicating slow convergence to stationarity under ill-conditioning.

A. Dual Perspective and KKT Conditions

Since the complexity of the matrix A directly impacts the difficulty of approximating $(A^\top A + L)^\dagger$, we adopt a dual formulation to decouple pseudoinverse from the specific realization of A . Introducing the auxiliary variable $y = Ax - b$, dual variables $\lambda \in \mathbb{R}^m$ with the constraint $y = Ax - b$ and $\mu \geq 0$ with the constraint $x \geq 0$, the resulting Lagrangian is:

$$\mathcal{L}(x, y; \lambda, \mu) = \frac{1}{2} \|y\|_2^2 + \frac{1}{2} x^\top Lx + \lambda^\top (Ax - b - y) - \mu^\top x. \quad (5)$$

KKT Conditions. Minimizing $\mathcal{L}$ with respect to y and x yields $y^* = \lambda$ and the stationarity condition $Lx + A^\top \lambda - \mu = 0$, respectively. Together with feasibility, and complementary slackness, the KKT conditions are:

$$x^* \succeq 0, \quad y^* = Ax^* - b, \quad \mu^* \succeq 0, \quad (6)$$

$$(\mu^*)^\top x^* = 0, \quad (7)$$

$$Lx^* + A^\top \lambda^* - \mu^* = 0. \quad (8)$$

Theorem 1. Assuming that $L^\dagger$ is known, from the KKT conditions (6)-(8), the primal solution of (3) can be denoted as:

$$x^* = L^\dagger(\mu^* - A^\top \lambda^*) + c\mathbf{1}, \quad c \in \mathbb{R}, \quad (9)$$

where μ^* and λ^* are dual solutions, and c must be determined to satisfy the KKT conditions.

Proof. From the KKT condition (8), we obtain:

$$Lx^* = \mu^* - A^\top \lambda^*. \quad (10)$$

To characterize the solution structure of (10), we recall fundamental properties of linear algebra [8], [40]. For any matrix $G \in \mathbb{R}^{m \times n}$ and equation $Gx = g$, the range space is $\mathcal{R}(G) = \{Gx \mid x \in \mathbb{R}^n\}$ and the null space is $\mathcal{N}(G) = \{x \mid Gx = 0\}$. The general solution, when it exists, takes the form $x = x_p + x_n$, where $Gx_p = g$ is a particular solution and $x_n \in \mathcal{N}(G)$ is an arbitrary null space component. Specifically:

- If G is invertible, $\mathcal{N}(G) = \{0\}$ and the unique solution is $x = G^{-1}g$.
- If G is singular but $g \in \mathcal{R}(G)$, the general solution is $x = G^\dagger g + x_n$ where $x_n \in \mathcal{N}(G)$ and $G^\dagger$ denotes the Moore–Penrose pseudoinverse.

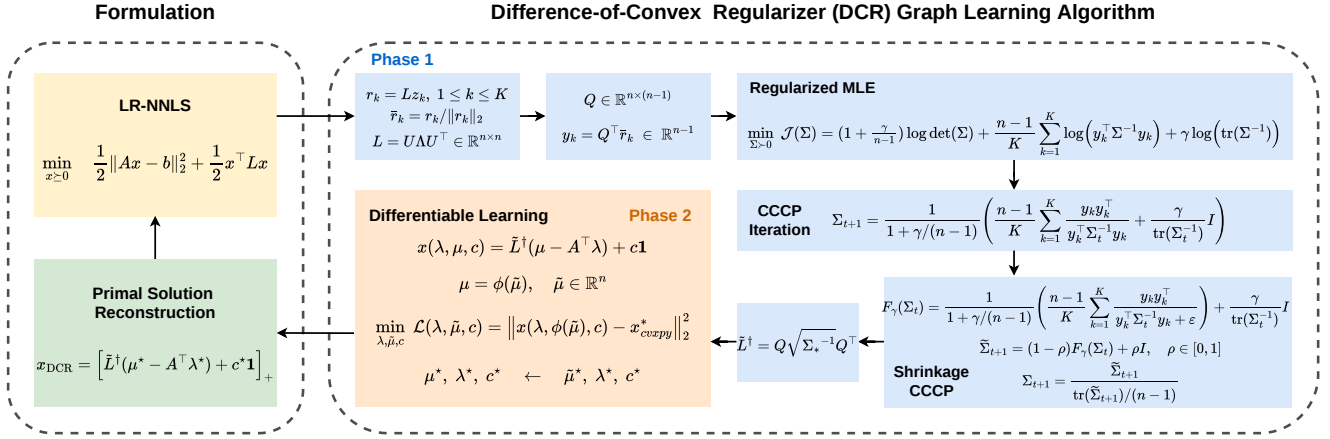


Fig. 3. DCR algorithm solves (3) in two phases. Phase 1 learns a Laplacian pseudoinverse approximation $\tilde{L}^\dagger$ via regularized MLE and CCCP iteration with shrinkage regularization, while Phase 2 recovers the primal solution through dual-driven differentiable learning.

The graph Laplacian L is symmetric positive semidefinite with $L\mathbf{1} = 0$, where $\mathbf{1} \in \mathbb{R}^n$ denotes the all-ones vector. For a connected graph, $\mathcal{N}(L) = \text{span}\{\mathbf{1}\}$ is one-dimensional, where “span” describes all the linear combinations of a set of vectors. Therefore, the right-hand side of (10) must lie in $\mathcal{R}(L)$, which is guaranteed by the complementary slackness of the KKT conditions. Thus, the general solution of (10) is:

$$x^* = L^\dagger(\mu^* - A^\top \lambda^*) + c\mathbf{1}, \quad c \in \mathbb{R}. \quad (11)$$

The constant c in (11) is uniquely determined by primal feasibility in (6) and complementary slackness (7). Let $\mathcal{I} = \{i \mid \mu_i^* > 0\}$ denote the active constraint set and $l = L^\dagger(\mu^* - A^\top \lambda^*)$. By complementary slackness, $x_i^* = 0$ for all $i \in \mathcal{I}$, which yields:

$$x_i^* = l_i + c = 0 \quad \Rightarrow \quad c = -l_i, \quad \forall i \in \mathcal{I}. \quad (12)$$

If $\mathcal{I} \neq \emptyset$, consistency of the KKT conditions requires l_i to be constant over $\mathcal{I}$, uniquely determining c ; the remaining components x_j^* ($j \notin \mathcal{I}$) satisfy the nonnegativity constraint. If $\mathcal{I} = \emptyset$ (i.e., $\mu^* = 0$), then $x^* \succ 0$ and c can be chosen arbitrarily without affecting the objective, typically $c = 0$. In both cases, (11) completely characterizes the optimal solution via $L^\dagger$ and the dual variables. $\square$

Example 1. To validate (11), we solve an LR-NNLS instance with $m = 50$ and $n = 30$. Given a connected graph Laplacian L , the primal problem is solved via CVXPY [41], yielding the primal solution x^* and dual solutions (λ^*, μ^*) . We reconstruct x^* from the dual solutions via: (i) computing c from active constraints (12), (ii) least-squares fitting $c = \frac{1}{n} \mathbf{1}^\top (x^* - l)$, and (iii) setting $c = 0$. Methods (i) and (ii) achieve $\mathcal{O}(10^{-6})$ error with nearly identical c . In contrast, (iii) yields a 62% relative error.

Objective. Given L , if $L^\dagger$ were available, the primal solution could be recovered via (11). However, computing $L^\dagger$ using eigendecomposition or matrix inversion is computationally prohibitive for large-scale systems. This motivates approximations of $L^\dagger$ on $\mathcal{R}(L)$.

IV. LAPLACIAN ESTIMATION

Since $L^\dagger$ is unknown, we propose the Difference-of-Convex Regularizer (DCR) graph learning framework to enable dual-guided primal solution recovery in (9). As shown in Algorithm 1, DCR consists of two phases (cf. Fig. 3). First, given L , DCR learns a matrix $\tilde{L}^\dagger$ that approximates the spectral action of $L^\dagger$, i.e., $\tilde{L}^\dagger \approx L^\dagger$, by solving a regularized MLE problem via the CCCP method [26]–[29]. Second, a differentiable reconstruction loss is introduced to recover the primal solution of (3) through gradient backpropagation using the high-accuracy reference solution x_{cvxpy}^* .

A. Regularized MLE for Laplacian Pseudoinverse Learning

Inspired by the Tyler’s estimator [42]–[45], this phase aims at formulating a scale-invariant regularized MLE problem [46] to learn $\tilde{L}^\dagger \in \mathbb{R}^{n \times n}$ for achieving $\tilde{L}^\dagger \approx L^\dagger$ on $\mathcal{R}(L)$.

1) *Tyler’s Estimator:* Tyler’s estimator stems from the Angular Central Gaussian (ACG) model, which describes scaled Gaussian vectors $x_i = \sqrt{\tau_i} \xi_i$ with $\xi_i \sim \mathcal{N}(0, \Sigma)$ and $\tau_i > 0$. To improve robustness over the sample covariance matrix $\hat{S} = \frac{1}{n} \sum_{i=1}^n x_i x_i^\top$, Tyler’s estimator identifies the matrix $\Sigma \in \mathbb{R}^{p \times p}$ via the fixed-point [43]:

$$\Sigma = \frac{p}{n} \sum_{i=1}^n \frac{x_i x_i^\top}{x_i^\top \Sigma^{-1} x_i}, \quad (13)$$

which is the stationarity condition of the ACG negative log-likelihood:

$$\ell(\Sigma) = \log \det(\Sigma) + \frac{p}{n} \sum_{i=1}^n \log(x_i^\top \Sigma^{-1} x_i). \quad (14)$$

Due to the scale invariance $\ell(c\Sigma) = \ell(\Sigma)$ ($c > 0$), a normalization such as $\text{tr}(\Sigma) = p$, is typically imposed to ensure solution uniqueness while preserving directional information.

TABLE I
REPRESENTATIVE REGULARIZERS FOR LAPLACIAN ESTIMATION.

Regularizer	Spectrum	Remarks
$\text{tr}(\Sigma^{-1})$	$\sum_i \lambda_i^{-1}$	Strong penalty on small eigenvalues
$\text{tr}(\Sigma^{-1/2})$	$\sum_i \lambda_i^{-1/2}$	Moderate spectral shrinkage
$\log(\text{tr}(\Sigma^{-1}))$	$\log(\sum_i \lambda_i^{-1})$	Scale-invariant and mild regularization
$-\log \det(\Sigma)$	$-\sum_i \log \lambda_i$	Log-barrier against eigenvalue collapse
$\ \Sigma^{-1}\ _*$	$\sum_i \lambda_i^{-1}$	Equivalent to $\text{tr}(\Sigma^{-1})$ for PSD matrices
$\ \Sigma^{-1/2}\ _*$	$\sum_i \lambda_i^{-1/2}$	Nuclear-norm form of spectral shrinkage

2) *Spectral Exploration*: In the absence of observed graph signals [34], we generate random directions to explore the spectrum of the Laplacian. For a connected graph, the Laplacian range space $\mathcal{R}(L)$ has dimension $n - 1$. Let $Q \in \mathbb{R}^{n \times (n-1)}$ be any orthonormal basis spanning $\mathcal{R}(L)$, e.g., the eigenvectors of L associated with its nonzero eigenvalues.

To cover all spectral directions, we sample isotropic Gaussian vectors $z_k \sim \mathcal{N}(0, I_n)$ and construct:

$$r_k = Lz_k, \quad \bar{r}_k = \frac{r_k}{\|r_k\|_2}, \quad k = 1, \dots, K. \quad (15)$$

Then $r_k \in \mathcal{R}(L)$ and hence $\bar{r}_k \in \mathcal{R}(L)$. With sufficiently large K , the set $\{r_k\}$ spans $\mathcal{R}(L)$ with high probability. The normalization removes magnitude variability while preserving directional information. Following the Tyler framework [42], [44], each normalized vector admits intrinsic coordinates:

$$y_k = Q^\top \bar{r}_k \in \mathbb{R}^{n-1}, \quad (16)$$

so that $\bar{r}_k = Qy_k$. The subsequent estimation is invariant to the choice of orthonormal basis Q .

3) *Regularized MLE on $\mathcal{R}(L)$* : To estimate $\tilde{L}^\dagger$ on $\mathcal{R}(L)$, we consider a regularized MLE formulation [45]:

$$\min_{\Sigma \succ 0} J(\Sigma) = (1 + \frac{\gamma}{n-1}) \log \det(\Sigma) + \frac{n-1}{K} \sum_{k=1}^K \log(y_k^\top \Sigma^{-1} y_k) + \gamma \mathcal{R}(\Sigma^{-1}), \quad (17)$$

where $\mathcal{R}(\cdot)$ is a spectral regularizer acting on the eigenvalues of Σ^{-1} . Such regularizers are commonly used to control the spectrum of Σ , in particular to prevent eigenvalue degeneracy, i.e., eigenvalues approaching zero. Representative choices are summarized in Table I (see Appendix A for details). In this work, we adopt the scale-invariant regularizer $\mathcal{R}(\Sigma^{-1}) = \log(\text{tr}(\Sigma^{-1}))$ to control the spectrum while preserving scale invariance. The resulting problem is:

$$\min_{\Sigma \succ 0} J(\Sigma) = (1 + \frac{\gamma}{n-1}) \log \det(\Sigma) + \frac{n-1}{K} \sum_{k=1}^K \log(y_k^\top \Sigma^{-1} y_k) + \gamma \log(\text{tr}(\Sigma^{-1})), \quad (18)$$

with $\gamma \geq 0$. To ensure identifiability under scale invariance, we impose $\text{tr}(\Sigma) = n - 1$.

Since $r_k = Lz_k$ with isotropic $z_k \sim \mathcal{N}(0, I_n)$, we have $\mathbb{E}[r_k r_k^\top] = L^2$. Although normalization removes scale variability, the directional structure remains aligned with L^2 on $\mathcal{R}(L)$. Hence, the optimal solution $\Sigma_\star$ of (18) shares eigenvectors with L and is spectrally aligned with L^2 , implying

Algorithm 1: DCR Algorithm

Input: A, b, L .

Output: $\tilde{L}^\dagger$ and x_{DCR} .

- 1 **Phase 1: Pseudoinverse Approximation Learning**
- 2 Compute an orthonormal basis $Q \in \mathbb{R}^{n \times (n-1)}$ of $\mathcal{R}(L)$; Sample $z_k \sim \mathcal{N}(0, I_n)$ and set $r_k = Lz_k$, $\bar{r}_k = r_k / \|r_k\|_2$; Compute $y_k = Q^\top \bar{r}_k \in \mathbb{R}^{n-1}$ for $k = 1, \dots, K$;
- 3 Initialize $\Sigma_1 \leftarrow I_{n-1}$;
- 4 **for** $t = 1, \dots, T_{\text{fp}}$ **do**
 - 5 $S \leftarrow \frac{n-1}{K} \sum_{k=1}^K \frac{y_k y_k^\top}{y_k^\top \Sigma_t^{-1} y_k + \varepsilon}$;
 - 6 $F_\gamma(\Sigma_t) \leftarrow \frac{1}{1 + \gamma/(n-1)} \left(S + \frac{\gamma}{\text{tr}(\Sigma_t^{-1})} I \right)$;
 - 7 $\tilde{\Sigma}_{t+1} \leftarrow (1 - \rho) F_\gamma(\Sigma_t) + \rho I$;
 - 8 $\Sigma_{t+1} \leftarrow \frac{\tilde{\Sigma}_{t+1}}{\text{tr}(\tilde{\Sigma}_{t+1})/(n-1)}$;
- 9 Compute $\tilde{L}^\dagger \leftarrow Q \sqrt{\Sigma_\star^{-1}} Q^\top$;
- 10 **Phase 2: Dual-Driven Differentiable Reconstruction**
- 11 Initialize auxiliary variables $(\lambda, \tilde{\mu}, c)$;
- 12 **for** $t = 1, \dots, T_1$ **do**
 - 13 $\mu \leftarrow \phi(\tilde{\mu})$;
 - 14 $x \leftarrow \tilde{L}^\dagger (\mu - A^\top \lambda) + c \mathbf{1}$;
 - 15 $\mathcal{L} \leftarrow \|x - x_{\text{cvxpy}}^*\|_2^2$;
 - 16 Update $(\lambda, \tilde{\mu}, c)$ by one gradient step on $\mathcal{L}$;
- 17 $c^* \leftarrow \arg \min_c \|[\tilde{L}^\dagger (\mu^* - A^\top \lambda^*) + c \mathbf{1}]_+ - x_{\text{cvxpy}}^*\|_2^2$;
- 18 **return** $x_{\text{DCR}} = [\tilde{L}^\dagger (\mu^* - A^\top \lambda^*) + c^* \mathbf{1}]_+$.

$\Sigma_\star^{-1} \propto (L^2)^\dagger$ on $\mathcal{R}(L)$. Let $\tilde{L}_{\mathcal{R}(L)}^\dagger = \sqrt{\Sigma_\star^{-1}}$, defined on $\mathcal{R}(L)$ and sharing eigenvectors with L . Then $\tilde{L}^\dagger$ is given by:

$$\tilde{L}^\dagger = Q \sqrt{\Sigma_\star^{-1}} Q^\top, \quad (19)$$

which serves as a learned approximation of $L^\dagger$.

B. CCCP Iteration

For the regularized objective (18), define $S_k \triangleq y_k y_k^\top$ and introduce $X \triangleq \Sigma^{-1} \succ 0$, then the problem becomes:

$$\min_{X \succ 0} F(X) = - (1 + \frac{\gamma}{n-1}) \log \det(X) + \frac{n-1}{K} \sum_{k=1}^K \log(\text{tr}(X S_k)) + \gamma \log(\text{tr}(X)). \quad (20)$$

Write $F(X) = f(X) + g(X)$ with:

$$f(X) = - (1 + \frac{\gamma}{n-1}) \log \det(X), \quad (21)$$

$$g(X) = \frac{n-1}{K} \sum_{k=1}^K \log(\text{tr}(X S_k)) + \gamma \log(\text{tr}(X)), \quad (22)$$

where $f(X)$ is convex and $g(X)$ is concave.

Defining $h(X) = -g(X)$, we can rewrite (20) to the Difference-of-Convex (DC) programming [18]–[21], [27]:

$$\min_{X \succ 0} F(X) = f(X) - h(X). \quad (23)$$

According to the definition of the convex conjugate, $h(X)$ admits the representation:

$$h(X) = \sup_Y \{\langle X, Y \rangle - h^*(Y)\}. \quad (24)$$

Substituting this representation to (23) yields:

$$f(X) - h(X) = \inf_Y \{f(X) - \langle X, Y \rangle + h^*(Y)\}, \quad (25)$$

which leads to the associated DC dual problem [18]:

$$\min_Y h^*(Y) - f^*(Y). \quad (26)$$

At an optimal solution X^* , the DC optimality condition [25], [27] requires:

$$\partial h(X^*) \subset \partial f(X^*), \quad (27)$$

which reduces to the stationarity condition:

$$\nabla f(X^*) = \nabla h(X^*), \quad (28)$$

when f and h are differentiable.

The CCCP method [26]–[28] provides an iterative scheme for the stationarity condition (28). Specifically, linearizing the concave part $g(X)$ in (22) at X_t yields the following convex subproblem:

$$X_{t+1} \in \arg \min_{X \succ 0} f(X) + \langle \nabla g(X_t), X \rangle. \quad (29)$$

The optimality condition gives:

$$\nabla f(X_{t+1}) = -\nabla g(X_t). \quad (30)$$

Since:

$$\nabla f(X) = -(1 + \frac{\gamma}{n-1})X^{-1}, \quad (31)$$

$$\nabla g(X) = \frac{n-1}{K} \sum_{k=1}^K \frac{S_k}{\text{tr}(XS_k)} + \frac{\gamma}{\text{tr}(X)}I, \quad (32)$$

we obtain:

$$(1 + \frac{\gamma}{n-1})X_{t+1}^{-1} = \frac{n-1}{K} \sum_{k=1}^K \frac{y_k y_k^\top}{y_k^\top X_t y_k} + \frac{\gamma}{\text{tr}(X_t)}I. \quad (33)$$

Using $\Sigma_t = X_t^{-1}$ yields the regularized fixed-point iteration:

$$\Sigma_{t+1} = \frac{1}{1 + \gamma/(n-1)} \left(\frac{n-1}{K} \sum_{k=1}^K \frac{y_k y_k^\top}{y_k^\top \Sigma_t^{-1} y_k} + \frac{\gamma}{\text{tr}(\Sigma_t^{-1})}I \right). \quad (34)$$

When $\gamma = 0$, the iteration reduces to (13).

By the standard CCCP descent property [26], i.e., $F(X_{t+1}) \leq F(X_t)$, we equivalently obtain:

$$\mathcal{J}(\Sigma_{t+1}) \leq \mathcal{J}(\Sigma_t), \quad \forall t \geq 0. \quad (35)$$

We solved (18) using three approaches: the CCCP iteration (34), a convex solver-based method, and the Disciplined Convex-Concave Programming (DCCP) package [19]. All three

methods converge to the same stationary point, providing numerical validation of the CCCP iteration.

Remark 1. Alternatively, problem (20) can also be solved using the Frank–Wolfe (FW) algorithm [47], [48], originally developed for minimizing convex quadratic functions over polytopes [49]. Specifically, recall the DC objective (23):

$$\min_{X \succ 0} F(X) = f(X) - h(X). \quad (36)$$

Introducing an epigraph variable yields:

$$\min_{X, y} y - h(X) \quad \text{s.t.} \quad f(X) \leq y. \quad (37)$$

Applying the FW algorithm, construct the Lagrangian function and the linear minimization step at X_t solves [49]:

$$\min_{X, y} y - \langle \nabla h(X_t), X \rangle \quad \text{s.t.} \quad f(X) \leq y, \quad (38)$$

then we can get $\nabla f(X_t^*) = \nabla h(X_t)$. Since:

$$\nabla f(X) = -(1 + \frac{\gamma}{n-1})X^{-1}, \quad (39)$$

$$\nabla h(X) = -\frac{n-1}{K} \sum_{k=1}^K \frac{y_k y_k^\top}{y_k^\top X y_k} - \frac{\gamma}{\text{tr}(X)}I, \quad (40)$$

we obtain:

$$(1 + \frac{\gamma}{n-1})X_{t+1}^{-1} = \frac{n-1}{K} \sum_{k=1}^K \frac{y_k y_k^\top}{y_k^\top X_t y_k} + \frac{\gamma}{\text{tr}(X_t)}I. \quad (41)$$

That is:

$$\Sigma_{t+1} = \frac{1}{1 + \gamma/(n-1)} \left(\frac{n-1}{K} \sum_{k=1}^K \frac{y_k y_k^\top}{y_k^\top \Sigma_t^{-1} y_k} + \frac{\gamma}{\text{tr}(\Sigma_t^{-1})}I \right), \quad (42)$$

which coincides with the iteration update (34). Thus, CCCP is equivalent to applying the FW algorithm for (20). In fact, FW-based methods have increasingly been used to solve related problems arising from CCCP frameworks.

Theorem 2. Let $\{y_k\}_{k=1}^K \subset \mathbb{R}^{n-1}$ satisfy $\sum_{k=1}^K y_k y_k^\top \succ 0$. For $\gamma \geq 0$, define the mapping $M_\gamma : \mathbb{S}_{++}^{n-1} \rightarrow \mathbb{S}_{++}^{n-1}$ by:

$$M_\gamma(\Sigma) = \frac{1}{1 + \gamma/(n-1)} \left(\frac{n-1}{K} \sum_{k=1}^K \frac{y_k y_k^\top}{y_k^\top \Sigma^{-1} y_k} + \frac{\gamma}{\text{tr}(\Sigma^{-1})}I \right). \quad (43)$$

Then M_γ is continuous, order-preserving, positively homogeneous of degree one, and strongly positive. Consequently, there exist $\Sigma_\star \succ 0$ and $\alpha_\star > 0$ such that $M_\gamma(\Sigma_\star) = \alpha_\star \Sigma_\star$. Moreover, the trace-normalized matrix $\hat{\Sigma}_\star \triangleq \frac{n-1}{\text{tr}(\Sigma_\star)} \Sigma_\star$ satisfies $\text{tr}(\hat{\Sigma}_\star) = n-1$ and is a fixed point of the normalized mapping:

$$\widehat{M}_\gamma(\Sigma) \triangleq \frac{(n-1) M_\gamma(\Sigma)}{\text{tr}(M_\gamma(\Sigma))}, \quad (44)$$

i.e., $\widehat{M}_\gamma(\hat{\Sigma}_\star) = \hat{\Sigma}_\star$.

Proof. Firstly, continuity follows from continuity of $\Sigma \mapsto \Sigma^{-1}$ on $\mathbb{S}_{++}^{n-1}$. Secondly, for $\alpha > 0$, using $(\alpha\Sigma)^{-1} = \alpha^{-1}\Sigma^{-1}$ gives:

$$M_\gamma(\alpha\Sigma) = \frac{1}{1 + \gamma/(n-1)} \left(\alpha \frac{n-1}{K} \sum_{k=1}^K \frac{y_k y_k^\top}{y_k^\top \Sigma^{-1} y_k} + \alpha \frac{\gamma}{\text{tr}(\Sigma^{-1})} I \right) = \alpha M_\gamma(\Sigma), \quad (45)$$

so M_γ is positively homogeneous. Meanwhile, if $\Sigma_1 \preceq \Sigma_2$, then $\Sigma_1^{-1} \succeq \Sigma_2^{-1}$, which implies:

$$\frac{y_k y_k^\top}{y_k^\top \Sigma_1^{-1} y_k} \preceq \frac{y_k y_k^\top}{y_k^\top \Sigma_2^{-1} y_k}, \quad (46)$$

for all k . Moreover, $\text{tr}(\Sigma_1^{-1}) \geq \text{tr}(\Sigma_2^{-1})$, which yields:

$$\frac{1}{\text{tr}(\Sigma_1^{-1})} I \preceq \frac{1}{\text{tr}(\Sigma_2^{-1})} I. \quad (47)$$

Hence $M_\gamma(\Sigma_1) \preceq M_\gamma(\Sigma_2)$ and T_γ is order-preserving. Additionally, since $y_k^\top \Sigma^{-1} y_k > 0$ and $I \succ 0$, the weighted sum defining $M_\gamma(\Sigma)$ is positive definite, implying strong positivity.

Finally, according to Nonlinear Perron–Frobenius (NPF) theory [50], [51], there exist $\Sigma_\star \succ 0$ and $\alpha_\star > 0$ such that $M_\gamma(\Sigma_\star) = \alpha_\star \Sigma_\star$. Let $\widehat{\Sigma}_\star \triangleq \frac{n-1}{\text{tr}(\Sigma_\star)} \Sigma_\star$ and define:

$$\widehat{M}_\gamma(\Sigma) = \frac{(n-1) M_\gamma(\Sigma)}{\text{tr}(M_\gamma(\Sigma))}. \quad (48)$$

By homogeneity, we can conclude $\widehat{M}_\gamma(\widehat{\Sigma}_\star) = \widehat{\Sigma}_\star$. $\square$

1) *Shrinkage-Stabilized CCCP Iteration:* Classical Tyler iterations are well defined when $K \geq n-1$, but may become ill-posed or unstable in undersampled regimes ($K < n-1$) [52]. While the regularized objective (18) improves spectral stability, additional shrinkage can further enhances robustness. Inspired by [53], we incorporate an isotropic shrinkage step. Given Σ_t , we first compute a stabilized update:

$$F_\gamma(\Sigma_t) = \frac{1}{1 + \gamma/(n-1)} \left(\frac{n-1}{K} \sum_{k=1}^K \frac{y_k y_k^\top}{y_k^\top \Sigma_t^{-1} y_k + \varepsilon} + \frac{\gamma}{\text{tr}(\Sigma_t^{-1})} I \right), \quad (49)$$

where $\varepsilon > 0$ is a small safeguard parameter for numerical stability. Shrinkage is then applied to the iterate:

$$\widetilde{\Sigma}_{t+1} = (1 - \rho) F_\gamma(\Sigma_t) + \rho I, \quad (50)$$

where $\rho \in [0, 1]$ is a shrinkage parameter used only for stabilization, while the primary spectral regularization is controlled by γ in (18). In practice, ρ is set to zero or a very small value when $K \geq n-1$, and activated when $K < n-1$.

Finally, trace normalization is applied:

$$\Sigma_{t+1} = \frac{\widetilde{\Sigma}_{t+1}}{\text{tr}(\widetilde{\Sigma}_{t+1})/(n-1)}. \quad (51)$$

When $\rho = 0$ and $\varepsilon = 0$, the iteration reduces to the regularized Tyler update (34), whose fixed point follows from NPF theory (Theorem 2). For $\rho \in (0, 1]$, although shrinkage breaks positive homogeneity of the mapping, a fixed point still exists as shown in Theorem 3.

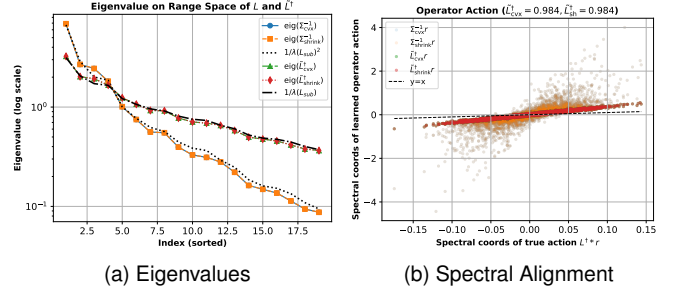


Fig. 4. MLE on $\mathcal{R}(L)$. (a) Σ^{-1} follows the theoretical eigenvalue scaling of $(L^2)^\dagger$ on $\mathcal{R}(L)$. (b): after $\widetilde{L}^\dagger = \sqrt{\Sigma^{-1}}$, $\widetilde{L}^\dagger$ aligns with $L^\dagger$ in spectral coordinates, confirming spectrum approximation of $L^\dagger$.

Theorem 3. Let $\varepsilon > 0$, $\rho \in (0, 1]$, and $\gamma \geq 0$. Assume $\sum_{k=1}^K y_k y_k^\top \succ 0$ and define:

$$F_\gamma(\Sigma) = \frac{1}{1 + \gamma/(n-1)} \left(\frac{n-1}{K} \sum_{k=1}^K \frac{y_k y_k^\top}{y_k^\top \Sigma^{-1} y_k + \varepsilon} + \frac{\gamma}{\text{tr}(\Sigma^{-1})} I \right), \quad (52)$$

$$\widetilde{M}(\Sigma) = (1 - \rho) F_\gamma(\Sigma) + \rho I, \quad (53)$$

$$M(\Sigma) = \frac{(n-1) \widetilde{M}(\Sigma)}{\text{tr}(\widetilde{M}(\Sigma))}. \quad (54)$$

Then the mapping M admits at least one fixed point $\Sigma_\star \succ 0$ with $\text{tr}(\Sigma_\star) = n-1$ such that $M(\Sigma_\star) = \Sigma_\star$.

Proof. Let $\mathcal{S} = \{\Sigma \succeq 0 : \text{tr}(\Sigma) = n-1\}$, and obviously, it is convex and compact. For any $\Sigma \succ 0$, there exists $y_k^\top \Sigma^{-1} y_k + \varepsilon > 0$, hence $F_\gamma(\Sigma)$ is well defined and bounded. Moreover, since $\rho \in (0, 1]$:

$$\widetilde{M}(\Sigma) = (1 - \rho) F_\gamma(\Sigma) + \rho I \succeq \rho I. \quad (55)$$

Thus $\widetilde{M}(\Sigma) \succ 0$. After trace normalization:

$$M(\Sigma) = \frac{(n-1) \widetilde{M}(\Sigma)}{\text{tr}(\widetilde{M}(\Sigma))} \succ 0, \quad \text{tr}(M(\Sigma)) = n-1, \quad (56)$$

which implies $M(\mathcal{S}) \subseteq \mathcal{S}$. Furthermore, M is continuous on $\mathcal{S}$ since all operations involved in $F_\gamma(\Sigma)$ and $\widetilde{M}(\Sigma)$ are continuous on $\mathbb{S}_{++}^{n-1}$. Therefore, by Brouwer's fixed-point theorem [54], there exists $\Sigma_\star \in \mathcal{S}$ such that $M(\Sigma_\star) = \Sigma_\star$. $\square$

Example 2. We evaluate the shrinkage-regularized CCCP iteration (49)–(51) for (18) with $\text{tr}(\Sigma) = n-1$ on a connected random graph ($n = 20$). We generate $K = 10n = 200$ samples $r_k = L z_k$, normalize $\bar{r}_k = r_k / \|r_k\|_2$, and obtain intrinsic coordinates $\{y_k\}_{k=1}^K$ on $\mathcal{R}(L)$. A CVXPY-based MM solver is used as a benchmark. Since $K \geq n-1$, the method reduces to a stabilized Tyler iteration ($\rho = 10^{-6}$) and converges to the same stationary point. The resulting Σ^{-1} closely matches the benchmark (normalized Frobenius inner product 1.0000, relative gap 9.7×10^{-5}). As shown in Fig. 4, Σ^{-1} aligns with $(L^2)^{-1}$ on $\mathcal{R}(L)$ (similarity ≈ 0.987), and $\widetilde{L}^\dagger = \sqrt{\Sigma^{-1}}$ matches $L^\dagger$ (similarity ≈ 0.992), consistent with theory (cf. Fig. 4a). Spectral tests (cf. Fig. 4b) further show strong agreement with $L^\dagger r$ (≈ 0.984), confirming accurate recovery on $\mathcal{R}(L)$.

C. Dual-Driven Differentiable Primal Reconstruction

In the second phase, the learned $\tilde{L}^\dagger$ is used to reconstruct a primal solution through a differentiable dual-driven parameterization learning progress. From the primal solution recovery formulation (9), we combine auxiliary dual variables $\lambda \in \mathbb{R}^m$, $\mu \in \mathbb{R}_+^n$, and a scalar bias $c \in \mathbb{R}$, and define the affine reconstruction map:

$$x(\lambda, \mu, c) = \tilde{L}^\dagger(\mu - A^\top \lambda) + c\mathbf{1}, \quad (57)$$

which is linear in (λ, μ, c) and therefore fully differentiable.

To remove the nonnegativity constraint $\mu \succeq 0$, we adopt a smooth reparameterization $\mu = \phi(\tilde{\mu})$ with $\tilde{\mu} \in \mathbb{R}^n$, where $\phi(\cdot)$ is the softplus function. This converts the constrained dual variables into unconstrained parameters and preserves end-to-end differentiability. With the $\tilde{L}^\dagger$ fixed from previous phase, we learn auxiliary dual variables that induce a primal candidate matching a high-accuracy reference solution x_{cvxpy}^* . Specifically, we solve the smooth differentiable training objective:

$$\min_{\lambda, \tilde{\mu}, c} \mathcal{L}(\lambda, \tilde{\mu}, c) = \|x(\lambda, \phi(\tilde{\mu}), c) - x_{\text{cvxpy}}^*\|_2^2. \quad (58)$$

During training, the nonnegativity projection on $x(\lambda, \mu, c)$ is omitted to preserve a fully differentiable computation graph. The variables $(\lambda, \tilde{\mu}, c)$ are jointly optimized under the smooth loss, where $\mu = \phi(\tilde{\mu})$ enforces nonnegativity. After convergence, $(\lambda^*, \tilde{\mu}^*)$ are fixed and $\mu^* = \phi(\tilde{\mu}^*)$ is obtained. The scalar bias c is then further corrected through a one-dimensional convex optimization that accounts for the final nonnegativity projection.

Finally, (λ^*, μ^*, c^*) are used to reconstruct the primal solution:

$$x_{\text{DCR}} = \left[\tilde{L}^\dagger(\mu^* - A^\top \lambda^*) + c^* \mathbf{1} \right]_+, \quad (59)$$

where $[\cdot]_+$ denotes the projection onto the nonnegative orthant.

V. NUMERICAL EXPERIMENTS

A. Experimental Setup

We evaluate DCR on the problem (3) with $n \in \{50, 100, 200, 500, 700, 1000, 1500, 2000\}$ and $m = 1.5n$. The matrix $A \in \mathbb{R}^{m \times n}$ has column-normalized i.i.d. standard Gaussian entries. Ground-truth signals $x_{\text{gt}} \in \mathbb{R}^n$ are 10% sparse with uniform nonzero values, and $b = Ax_{\text{gt}}$. All experiments are implemented in PyTorch with GPU acceleration.

DCR first learns a symmetric approximation $\tilde{L}^\dagger \approx L^\dagger$ via fixed-point iterations (49)–(51) using an adaptive number of random samples:

$$K(n) = \min \left\{ K_{\max}, \max \{ K_{\min}, \text{round}(2\sqrt{n}) \} \right\}. \quad (60)$$

To improve stability, the shrinkage weight ρ in (50) is selected adaptively:

$$\rho = \text{clip} \left(\hat{\rho}_{\text{plugin}} \cdot \max \left\{ 0, 1 - \frac{K}{n-1} \right\}, \frac{1}{1 + c_\gamma \gamma}, \rho_{\min}, \rho_{\max} \right), \quad (61)$$

where $\hat{\rho}_{\text{plugin}}$ is a data-driven estimate, K controls the number of samples, $(1 + c_\gamma \gamma)^{-1}$ attenuates shrinkage as the regularization parameter γ increases, and $[\rho_{\min}, \rho_{\max}]$ defines the admissible range.

TABLE II
EXPERIMENTAL PARAMETERS.

Parameter	Value
Problem size n	$\{50, 100, 200, 500, 700, 1000, 1500, 2000\}$
Measurements m	$1.5n$
Laplacian topology	grid2d, Erdős–Rényi (ER)
Grid weight jitter	0.2
Phase 1 iterations	$\min\{3000, 500 + 2(n-1)\}$
Residual samples K	$\min(K_{\max}, \max(K_{\min}, \lceil 2\sqrt{n} \rceil))$
Shrinkage ρ	Adaptive
Regularization (ϵ, δ)	$(10^{-6}, 10^{-3})$
Reference solver	CVXPY (OSQP / SCS)
Solver tolerance	10^{-5}
Max solver iterations	2×10^5
Phase 2 iterations	30000
Dual learning rate η_D	5×10^{-3}
Softplus temperature β	0.5

High-accuracy reference solutions are obtained via the convex solver CVXPY [41]. The dual variables (λ, μ, c) are optimized using Adam with Softplus reparameterization ($\mu \succeq 0$), and the primal solution is reconstructed via (59). All hyperparameters are listed in Table II.

1) *Baselines*: We benchmark DCR against the following methods:

- CVXPY [41]: a convex optimization solver that directly solves (3) using OSQP or SCS, providing high-accuracy reference solutions.
- Chebyshev polynomial approximation [11]–[13]: the action of the Laplacian pseudoinverse $L^\dagger$ on a vector v is approximated via a degree- K_{cb} Chebyshev polynomial expansion of the function $1/\lambda$ over the nonzero spectral interval $[\lambda_2(L), \lambda_{\max}(L)]$, where $\lambda_2(L)$ and $\lambda_{\max}(L)$ denote the second-smallest and largest eigenvalues of L , respectively. Specifically, $L^\dagger v \approx \sum_{k=0}^{K_{cb}} \alpha_k T_k(\tilde{L}) v$, where $T_k(\cdot)$ denotes the Chebyshev polynomial and $\tilde{L}$ is the rescaled Laplacian whose spectrum is linearly mapped to $[-1, 1]$. This approximation enables a matrix-free implementation of inverse filtering and is integrated into a gradient-based iterative solver for (3).

2) *Evaluation Metrics*: Performance of all methods is quantified using the following metrics:

- Solution reconstruction error, including:

$$\text{Max Abs. Err.} := \max_i |x_i - x_i^{\text{CVXPY}}|, \quad (62)$$

$$\text{Mean Abs. Err.} := \frac{1}{n} \sum_i |x_i - x_i^{\text{CVXPY}}|, \quad (63)$$

$$\text{Rel. Sol. Err.} := \frac{\|x - x_{\text{CVXPY}}\|_2}{\|x_{\text{CVXPY}}\|_2}. \quad (64)$$

- Relative objective value gap:

$$\text{Rel. Obj. Gap} := \frac{|Y - Y_{\text{CVXPY}}|}{|Y_{\text{CVXPY}}|}, \quad (65)$$

where Y and Y_{CVXPY} denote the objective values of (3) evaluated at the solutions obtained by the compared methods and by CVXPY, respectively.

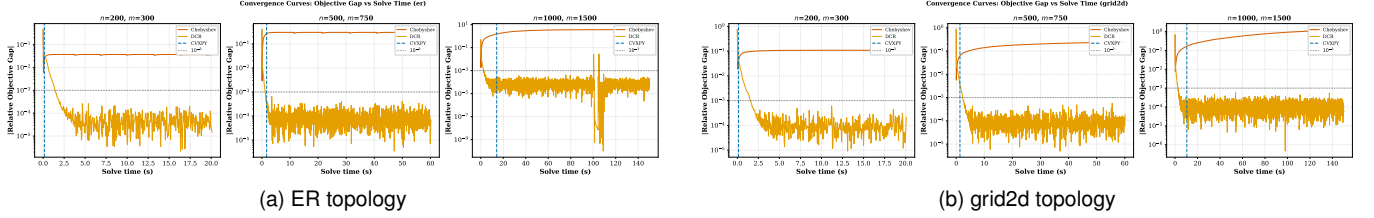


Fig. 5. Convergence curves (objective gap vs. solve time) for DCR and Chebyshev on the LR-NNLS problem (3) under two graph topologies, at $n \in \{200, 500, 1000\}$ with $m = 1.5n$. Each subplot shows the absolute relative objective gap (65) versus solve time; the vertical dashed line marks the CVXPY solve time; the horizontal dotted line marks the 10^{-3} accuracy threshold. DCR consistently achieves rapid convergence to high-accuracy solutions (typically below 10^{-4}), while Chebyshev exhibits slow convergence and stagnates at significantly higher objective gaps, especially for larger problem sizes.

- Runtime (time-to-accuracy): the time required for a method to first satisfy:

$$\left| \frac{Y - Y_{\text{CVXPY}}}{Y_{\text{CVXPY}}} \right| \leq 10^{-3}. \quad (66)$$

For DCR, only Phase 2 (online) time is counted; Phase 1 is excluded as it is amortized across instances with the same graph.

- Topology robustness: experiments are conducted under both grid2d and Erdős–Rényi (ER) topologies, which exhibit markedly different spectral gaps and conditioning characteristics, to assess sensitivity to graph structure.

B. Numerical Results

1) *Convergence Behavior*: Fig. 5 shows the convergence behavior of DCR and Chebyshev in terms of relative objective gap versus solve time under both grid2d and ER topologies.

Grid2d Topology. Fig. 5 (a) shows the convergence results on grid2d graphs. DCR converges rapidly for all problem sizes: for $n = 200$, the objective gap drops below 10^{-3} within 2.47 s; for $n = 500$ and $n = 1000$, the time-to-accuracy is 2.27 s and 2.76 s, respectively. After the initial fast decrease, the objective gap stabilizes, staying around 10^{-5} to 10^{-3} for $n = 200$ and 10^{-5} to 10^{-4} for $n = 500$, with small variations due to the Adam optimizer. For $n = 1000$, DCR achieves a final relative objective gap of 5.9×10^{-11} and a relative solution error of 3.3×10^{-9} , which are close to machine precision. In contrast, Chebyshev shows poor convergence: for $n = 200$ and $n = 500$, the objective gap increases after the first few iterations and then stabilizes around 10^{-1} , while for $n = 1000$ it remains above 10^{-1} throughout the entire time budget. In all cases, Chebyshev fails to reach the 10^{-3} threshold, indicating that the fixed polynomial approximation introduces a bias that cannot be reduced by further iterations.

ER Topology. Fig. 5 (b) shows that DCR maintains similar convergence behavior on ER graphs despite their different spectral properties. For $n = 200$, DCR reaches the 10^{-3} threshold at 1.85 s and then converges to a gap around 10^{-4} – 10^{-5} ; for $n = 500$, it reaches the threshold at 1.54 s and stabilizes at a similar level; for $n = 1000$, it crosses the threshold at 3.19 s and achieves a final gap of 2.3×10^{-11} . The corresponding CVXPY solve times are 0.16 s, 1.67 s, and 14.14 s, showing that DCR remains efficient as the problem size grows. Chebyshev performs even worse on ER graphs: for example, at $n = 500$ and $n = 700$, the relative solution

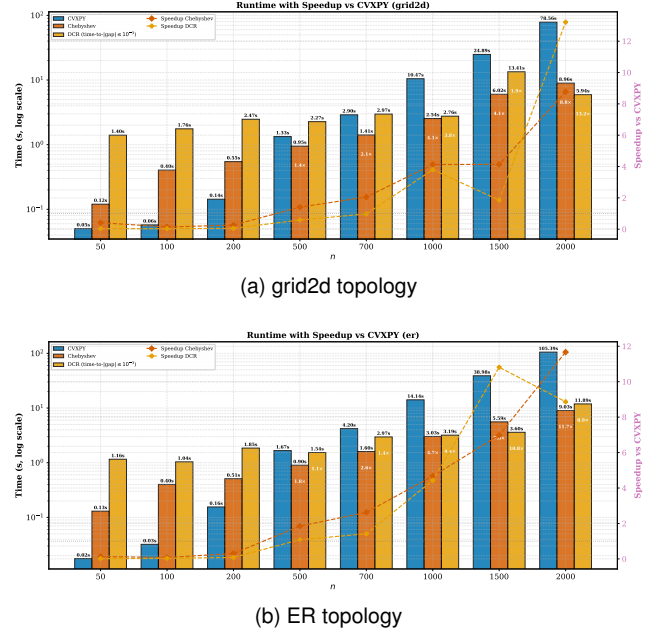


Fig. 6. Runtime and speedup comparison of DCR and Chebyshev relative to CVXPY on the LR-NNLS problem (3) for $n \in [50, 2000]$, $m = 1.5n$. (a): grid2d topology. (b): ER topology. Bar heights show runtime. White labels inside Chebyshev and DCR bars indicate the speedup ratio relative to CVXPY. DCR runtime is measured as time-to-accuracy (66). The right axis (dashed lines) shows the speedup ratio of each method relative to CVXPY.

error reaches 1.20 and 1.17, which is significantly larger than the grid2d case (0.36 and 0.74), indicating that the polynomial approximation of $L^\dagger$ degrades under more irregular spectra. In contrast, DCR maintains stable accuracy across both topologies, with final objective gaps on ER graphs (7.9×10^{-10} , 8.2×10^{-10} , and 2.3×10^{-11} for $n = 500, 700, 1000$) comparable to those on grid2d, demonstrating that the learned approximation $\tilde{L}^\dagger$ in DCR adapts well to different graph structures without modifying the algorithm.

2) *Runtime and Speedup Analysis*: Fig. 6 compares the runtime and speedup of DCR and Chebyshev relative to CVXPY under both grid2d and ER topologies.

Grid2d Topology. Fig. 6 (a) shows the runtime on grid2d graphs. CVXPY's runtime increases rapidly with problem size, from 0.05 s at $n = 50$ to 78.56 s at $n = 2000$, indicating poor scaling with problem size. Chebyshev is faster at small sizes (e.g., 0.12 s at $n = 50$), but its runtime still grows steadily and reaches 8.96 s at $n = 2000$. Moreover, its speedup

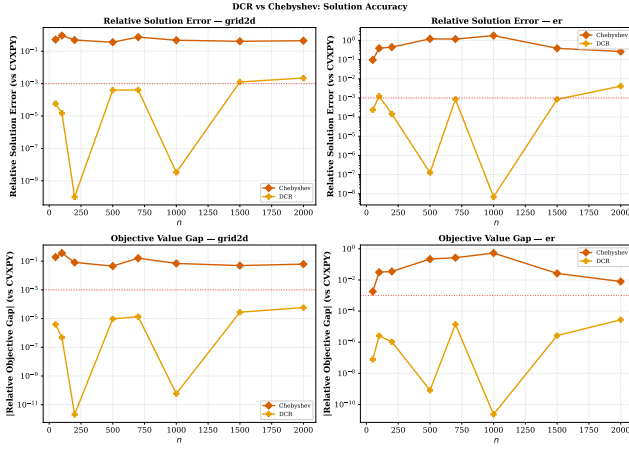


Fig. 7. Solution accuracy of DCR and Chebyshev on the LR-NNLS problem (3) for $n \in [50, 2000]$, $m = 1.5n$, under grid2d (left column) and ER (right column) topologies. Top row: relative solution error (64) versus n . Bottom row: relative objective gap (65) versus n . The red dotted line marks the 10^{-3} threshold. All metrics are computed relative to the convex solver reference solution. DCR consistently achieves high accuracy (often below 10^{-5}), while Chebyshev exhibits large errors and fails to reach the desired accuracy across all problem sizes, especially on ER graphs.

over CVXPY remains limited, below $4.1\times$ for $n \leq 1000$ and peaking at $8.8\times$ at $n = 2000$, while its low solution accuracy (as shown in Fig. 5 (b)) reduces the practical value of this speed advantage. In contrast, DCR shows stable runtime across a wide range of problem sizes. It requires 1.40 s at $n = 50$, 2.27 s at $n = 500$, 2.76 s at $n = 1000$, and 5.94 s at $n = 2000$ to reach the target accuracy. As a result, the speedup of DCR increases with n , reaching $3.8\times$ at $n = 1000$ and $13.2\times$ at $n = 2000$. These results show that DCR scales well and becomes more efficient than the convex solver as the problem size grows.

ER Topology. Fig. 6 (b) shows the runtime on ER graphs. CVXPY becomes more expensive as n increases, rising from 0.02 s at $n = 50$ to 105.39 s at $n = 2000$. DCR maintains stable time-to-accuracy across scales, requiring 1.16 s at $n = 50$, 1.54 s at $n = 500$, 3.19 s at $n = 1000$, 3.60 s at $n = 1500$, and 11.89 s at $n = 2000$. This leads to increasing speedup over the convex solver, reaching $4.4\times$ at $n = 1000$, $10.8\times$ at $n = 1500$, and $8.9\times$ at $n = 2000$. Chebyshev achieves a higher speedup at the largest size (up to $11.7\times$ at $n = 2000$), but this comes at the cost of poor solution quality. As shown in Fig. 5 (a), Chebyshev does not meet the 10^{-3} accuracy threshold at any scale on ER graphs. Thus, its apparent runtime advantage is limited when solution accuracy is taken into account. DCR achieves consistent and increasing speedup with problem size while maintaining high solution accuracy, offering a better efficiency–accuracy trade-off across different graph structures.

3) Solution Accuracy: Fig. 7 compares the solution accuracy of DCR and Chebyshev in terms of relative solution error and objective gap with respect to the CVXPY reference.

Grid2d Topology. Fig. 7 (left column) shows the results on grid2d graphs. DCR consistently achieves high accuracy across all problem sizes, with relative solution error typically below 10^{-3} and often reaching 10^{-5} to 10^{-10} . For example, the relative solution error drops to 1.0×10^{-10} at $n = 200$ and

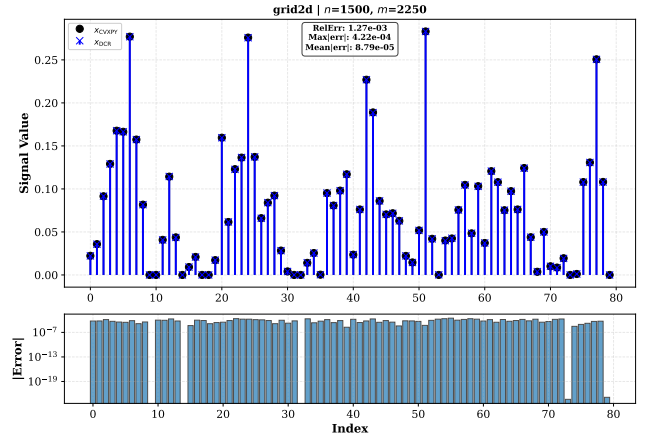


Fig. 8. Signal reconstruction comparison between DCR and CVXPY on a grid2d instance ($n = 1500$, $m = 2250$). Top: first 80 components of x_{CVXPY} (black) and x_{DCR} (blue), which are visually indistinguishable. Bottom: elementwise absolute error $|x_{\text{DCR}} - x_{\text{CVXPY}}|$ (log scale), concentrated around 10^{-7} – 10^{-6} . Overall statistics: RelErr = 1.27×10^{-3} (64), Max |err| = 4.22×10^{-4} (62), Mean |err| = 8.79×10^{-5} (63).

3.3×10^{-9} at $n = 1000$. The corresponding objective gap is also extremely small, reaching 2.1×10^{-12} at $n = 200$ and 5.9×10^{-11} at $n = 1000$. In contrast, Chebyshev exhibits large errors across all scales, with relative solution error around 0.3 – 0.9 and objective gap around 10^{-2} to 10^{-1} , and it never approaches the 10^{-3} threshold. This suggests that the fixed polynomial approximation introduces a bias that is difficult to reduce by increasing iterations.

ER Topology. Fig. 7 (right column) shows similar trends on ER graphs. DCR again maintains high accuracy, with relative solution error typically below 10^{-3} and reaching as low as 1.3×10^{-7} at $n = 500$ and 6.8×10^{-9} at $n = 1000$. The objective gap follows the same pattern, achieving values as small as 8.2×10^{-10} and 2.3×10^{-11} . In contrast, Chebyshev performs worse on ER graphs than on grid2d graphs, with relative solution error exceeding 1 at several scales, reaching 1.20 at $n = 500$ and 1.78 at $n = 1000$, and with objective gap remaining above 10^{-2} in most cases. These results suggest that the polynomial approximation of $L^\dagger$ may degrade under more irregular spectra, while DCR remains accurate and stable across different graph structures.

4) Signal Reconstruction Accuracy: Fig. 8 shows a pointwise comparison between x_{DCR} and the convex solver reference solution for a grid2d instance with $n = 1500$ and $m = 2250$. The two signals are visually indistinguishable across all displayed components, indicating that DCR accurately recovers both support and magnitude. The elementwise absolute solution error remains uniformly small, typically in the range 10^{-8} to 10^{-6} , with no noticeable outliers or structured deviations. Quantitatively, the relative solution error is 1.27×10^{-3} , with maximum and mean absolute solution errors of 4.22×10^{-4} and 8.79×10^{-5} , respectively. The relative objective gap is 2.78×10^{-5} , confirming that DCR achieves a solution very close to the convex solver reference solution. These results show that DCR provides accurate and stable reconstruction at large problem scale.

TABLE III
PERFORMANCE SUMMARY OF DCR AND CHEBYSHEV ($m = 1.5n$, CVXPY AS REFERENCE). DCR RUNTIME IS TIME-TO-ACCURACY. SPEEDUP IS RELATIVE TO THE CONVEX SOLVER CVXPY. BEST RESULTS PER ROW IN **BOLD**.

Topology	n	CVXPY Time (s)	Chebyshev		DCR (Ours)			
			Rel. Sol. Err.	Rel. Obj. Gap	Rel. Sol. Err.	Rel. Obj. Gap	Time (s)	Speedup
grid2d	50	0.05	5.26×10^{-1}	1.92×10^{-1}	5.79×10^{-5}	3.93×10^{-6}	1.40	$0.04 \times$
	100	0.06	9.22×10^{-1}	3.66×10^{-1}	1.51×10^{-5}	4.81×10^{-7}	1.76	$0.03 \times$
	200	0.14	4.91×10^{-1}	8.17×10^{-2}	1.02×10^{-10}	2.13×10^{-12}	2.47	$0.06 \times$
	500	1.33	3.61×10^{-1}	4.54×10^{-2}	3.96×10^{-4}	9.48×10^{-6}	2.27	$0.6 \times$
	700	2.90	7.44×10^{-1}	1.62×10^{-1}	4.08×10^{-4}	1.35×10^{-5}	2.97	$1.0 \times$
	1000	10.47	4.77×10^{-1}	6.94×10^{-2}	3.32×10^{-9}	5.87×10^{-11}	2.76	$3.8 \times$
	1500	24.89	4.11×10^{-1}	4.93×10^{-2}	1.27×10^{-3}	2.78×10^{-5}	13.41	$1.9 \times$
	2000	78.56	4.47×10^{-1}	6.24×10^{-2}	2.22×10^{-3}	5.83×10^{-5}	5.94	$13.2 \times$
ER	50	0.02	9.63×10^{-2}	1.82×10^{-3}	2.39×10^{-4}	7.85×10^{-8}	1.16	$0.02 \times$
	100	0.03	3.91×10^{-1}	3.15×10^{-2}	1.22×10^{-3}	2.59×10^{-6}	1.04	$0.03 \times$
	200	0.16	4.46×10^{-1}	3.51×10^{-2}	1.45×10^{-4}	1.07×10^{-6}	1.85	$0.08 \times$
	500	1.67	1.20×10^0	2.18×10^{-1}	1.25×10^{-7}	8.17×10^{-10}	1.54	$1.1 \times$
	700	4.20	1.17×10^0	2.67×10^{-1}	8.45×10^{-4}	1.41×10^{-5}	2.97	$1.4 \times$
	1000	14.14	1.78×10^0	5.32×10^{-1}	6.85×10^{-9}	2.32×10^{-11}	3.19	$4.4 \times$
	1500	38.98	3.88×10^{-1}	2.63×10^{-2}	8.43×10^{-4}	2.68×10^{-6}	3.60	$10.8 \times$
	2000	105.39	2.62×10^{-1}	7.92×10^{-3}	4.09×10^{-3}	2.74×10^{-5}	11.89	$8.9 \times$

5) *Summary*: Table III summarizes the performance of DCR and Chebyshev across all configurations. First, DCR consistently achieves much higher accuracy, outperforming Chebyshev by several orders of magnitude on both grid2d and ER topologies. While Chebyshev’s relative solution error remains above 9.6×10^{-2} and even exceeds 1 on ER graphs, DCR attains errors as low as 10^{-10} on grid2d and 10^{-9} on ER, with almost all cases satisfying the 10^{-3} criterion. Second, the speedup of DCR over the convex solver increases with problem size. Although DCR is slower at small scales, it achieves up to $13.2 \times$ speedup on grid2d graph and $10.8 \times$ on ER graph at large n , demonstrating that DCR becomes increasingly competitive as problem size grows. Third, DCR maintains both accuracy and efficiency across different graph topologies, indicating that the learned pseudoinverse approximation $\tilde{L}^\dagger$ is robust to variations in graph structure.

VI. CONCLUSION

This work addresses a key challenge in Laplacian-regularized optimization caused by the dense and ill-conditioned Laplacian pseudoinverse. We propose a Difference-of-Convex Regularizer (DCR) graph learning framework that replaces direct pseudoinverse computation with a differentiable, spectrally structured approximation learned via regularized Maximum Likelihood Estimation (MLE) and Convex-Concave Procedure (CCCP) iterations. By reformulating Laplacian-Regularized Nonnegative Least Squares (LR-NNLS) through a dual representation, DCR decouples pseudoinverse approximation from instance-specific optimization, enabling cross-instance reuse and efficient primal recovery. We further establish theoretical guarantees on fixed-point existence and stability. Experiments across diverse graph topologies and scales show that DCR captures the inverse spectral behavior of the Laplacian, achieving stable convergence and accurate reconstruction under severe

ill-conditioning. Compared with convex solvers and graph filtering methods, DCR achieves similar accuracy while significantly improving performance at large scales. Future work will extend the framework to broader graph-regularized problems and dynamic graph settings.

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APPENDIX

This appendix derives CCCP iterations for two representative spectral regularizers in Table I. As in the main text, we set $X = \Sigma^{-1}$ and formulate the objective as a DC program, with $S_k = y_k y_k^T$. Unlike the scale-invariant regularization in IV-A, these variants alter the scaling behavior of the Tyler likelihood. To ensure well-posedness and avoid pathological scaling, we enforce the normalization $\text{tr}(\Sigma) = n - 1$.

A. Nuclear-Norm Regularization: $\|\Sigma^{-1/2}\|_*$

Consider the following nuclear-norm regularized MLE:

$$\min_{\Sigma \succ 0} J(\Sigma) = \log \det(\Sigma) + \frac{n-1}{K} \sum_{k=1}^K \log(y_k^T \Sigma^{-1} y_k) + \gamma \|\Sigma^{-1/2}\|_*. \quad (67)$$

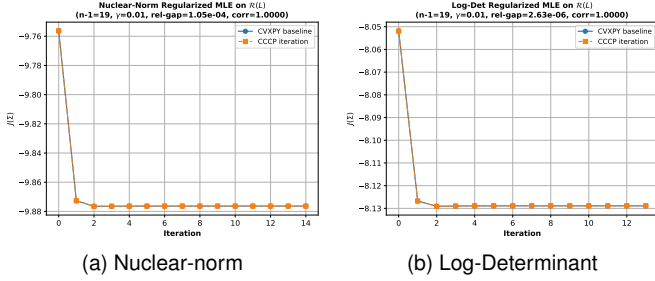


Fig. 9. Convergence comparison between the CVXPY baseline and the fixed-point iterations (75) and (83) for (67) and (76) on $\mathcal{R}(L)$. Nearly identical objective trajectories confirm consistency with the CCCP formulation.

Here, the regularizer $\gamma \|\Sigma^{-1/2}\|_*$ penalizes small eigenvalues of Σ while inducing a milder spectral shrinkage compared with $\text{tr}(\Sigma^{-1})$. Note that for positive definite matrices Σ :

$$\|\Sigma^{-1/2}\|_* = \text{tr}(\sqrt{(\Sigma^{-1/2})^\top (\Sigma^{-1/2})}) = \sum_i \lambda_i^{-1/2}. \quad (68)$$

Then the problem (67) becomes:

$$\begin{aligned} \min_{\Sigma \succ 0} J(\Sigma) = & \log \det(\Sigma) + \frac{n-1}{K} \sum_{k=1}^K \log(y_k^\top \Sigma^{-1} y_k) \\ & + \gamma \text{tr}(\Sigma^{-1/2}). \end{aligned} \quad (69)$$

CCCP Iteration. Let $X = \Sigma^{-1} \succ 0$. Since $\text{tr}(\Sigma^{-1/2}) = \text{tr}(X^{1/2})$, the objective becomes:

$$\begin{aligned} F(X) = & -\log \det(X) + \frac{n-1}{K} \sum_{k=1}^K \log(\text{tr}(X S_k)) \\ & + \gamma \text{tr}(X^{1/2}). \end{aligned} \quad (70)$$

Define $f(X) = -\log \det(X)$ and:

$$g(X) = \frac{n-1}{K} \sum_{k=1}^K \log(\text{tr}(X S_k)) + \gamma \text{tr}(X^{1/2}), \quad (71)$$

where $f(X)$ is convex on $X \succ 0$, while $g(X)$ is concave. Linearizing the concave function $g(X)$ at X_t gives:

$$X_{t+1} \in \arg \min_{X \succ 0} f(X) + \langle \nabla g(X_t), X \rangle. \quad (72)$$

The gradients are $\nabla f(X) = -X^{-1}$ and:

$$\nabla g(X) = \frac{n-1}{K} \sum_{k=1}^K \frac{S_k}{\text{tr}(X S_k)} + \frac{\gamma}{2} X^{-1/2}. \quad (73)$$

The optimality condition of (72) yields:

$$X_{t+1}^{-1} = \frac{n-1}{K} \sum_{k=1}^K \frac{y_k y_k^\top}{y_k^\top X_t y_k} + \frac{\gamma}{2} X_t^{-1/2}. \quad (74)$$

Using $\Sigma_t = X_t^{-1}$, we obtain:

$$\Sigma_{t+1} = \frac{n-1}{K} \sum_{k=1}^K \frac{y_k y_k^\top}{y_k^\top \Sigma_t^{-1} y_k} + \frac{\gamma}{2} \Sigma_t^{1/2}, \quad (75)$$

followed by the normalization $\text{tr}(\Sigma_{t+1}) = n-1$. When $\gamma = 0$, the iteration reduces to the classical Tyler update.

Validation. Fig. 9 (a) compares CVXPY with the fixed-point iteration (75) for (67) on $\mathcal{R}(L)$. The trajectories nearly coincide and converge within a few iterations, confirming the correctness of the CCCP update, with relative Frobenius gap 1.05×10^{-4} and correlation 1.0000.

B. Log-Determinant Barrier Regularization: $-\gamma \log \det(\Sigma)$

Consider the following log-determinant regularized MLE:

$$\begin{aligned} \min_{\Sigma \succ 0} J(\Sigma) = & \log \det(\Sigma) + \frac{n-1}{K} \sum_{k=1}^K \log(y_k^\top \Sigma^{-1} y_k) \\ & - \gamma \log \det(\Sigma). \end{aligned} \quad (76)$$

The regularizer $-\gamma \log \det(\Sigma)$ acts as a spectral barrier that prevents eigenvalues of Σ from collapsing to zero. Since $-\log \det(\Sigma) = -\sum_i \log \lambda_i$, this penalty grows unbounded as any eigenvalue $\lambda_i \rightarrow 0$, thereby stabilizing the spectrum of Σ . Combining the log-determinant terms, problem (76) becomes:

$$\min_{\Sigma \succ 0} J(\Sigma) = (1-\gamma) \log \det(\Sigma) + \frac{n-1}{K} \sum_{k=1}^K \log(y_k^\top \Sigma^{-1} y_k). \quad (77)$$

CCCP Iteration. Let $X = \Sigma^{-1} \succ 0$, and (77) becomes:

$$F(X) = -(1-\gamma) \log \det(X) + \frac{n-1}{K} \sum_{k=1}^K \log(\text{tr}(X S_k)). \quad (78)$$

Define $f(X) = -(1-\gamma) \log \det(X)$ and:

$$g(X) = \frac{n-1}{K} \sum_{k=1}^K \log(\text{tr}(X S_k)), \quad (79)$$

where $f(X)$ is convex on $X \succ 0$, while $g(X)$ is concave. Linearizing $g(X)$ at X_t yields:

$$X_{t+1} \in \arg \min_{X \succ 0} f(X) + \langle \nabla g(X_t), X \rangle. \quad (80)$$

The gradients are $\nabla f(X) = -(1-\gamma)X^{-1}$ and:

$$\nabla g(X) = \frac{n-1}{K} \sum_{k=1}^K \frac{S_k}{\text{tr}(X S_k)}. \quad (81)$$

The optimality condition of (80) gives:

$$(1-\gamma)X_{t+1}^{-1} = \frac{n-1}{K} \sum_{k=1}^K \frac{y_k y_k^\top}{y_k^\top X_t y_k}. \quad (82)$$

Using $\Sigma_t = X_t^{-1}$, we obtain:

$$\Sigma_{t+1} = \frac{1}{1-\gamma} \left(\frac{n-1}{K} \sum_{k=1}^K \frac{y_k y_k^\top}{y_k^\top \Sigma_t^{-1} y_k} \right), \quad (83)$$

followed by the normalization $\text{tr}(\Sigma_{t+1}) = n-1$. When $\gamma = 0$, the iteration reduces to the classical Tyler fixed-point update.

Validation. Fig. 9 (b) shows the convergence of CVXPY and the fixed-point iteration (83) for (76) on $\mathcal{R}(L)$. The two trajectories almost overlap, confirming the correctness of the CCCP update. Both methods converge within a few iterations and yield nearly identical solutions, with a relative Frobenius gap of 2.63×10^{-6} and correlation 1.0000.